

ARKA GHOSH

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EDUCATION

George Mason University, Fairfax, VA, USA <i>PhD Information Science and Technology</i>	2024 - Present 4.0/4.0 GPA
Digital University Kerala (Formerly IIITM-K), India <i>M.Sc. Computer Science with specialization in Cybersecurity</i>	2022 - 2024 8.44/10 CGPA (2nd in Class)
University of Calcutta, India <i>B.Sc. Computer Science</i>	2019 - 2022 8.71/10 CGPA (5th in class)

EXPERIENCES

Internships:

- **AI Cybersecurity Product Development Researcher - Akiak Technology (May 2026-Aug 2026)**
 - Developing an AI-assisted cybersecurity platform for automating NIST Risk Management Framework (RMF) and Authorization to Operate (ATO) workflows.
 - Designing automated generation and continuous synchronization of System Security Plans (SSPs) with evolving system configurations and security controls.
 - Developing continuous monitoring capabilities to identify configuration and compliance changes and support continuous ATO (cATO).
 - Exploring AI-driven security control assessment and compliance automation to reduce manual effort across the system authorization lifecycle.
- **Cybersecurity Field Training Experience, CCI (Sept 2025-Present)**
 - Leading a project on attack path construction and design flaw detection using CVEs.
 - Conducting field-oriented cybersecurity research bridging academic innovation with defense applications.
 - Guiding two undergraduate researchers in experimental development and analysis.
- **Graduate Research Assistant, George Mason University(June 2025-August 2025)**
 - Developed an AI-powered tutoring agent for ethics training for an undergraduate course.
 - Applied RAG for context-specific feedback by the agent for the students.
 - Deployed the AI agent on GMU's internal server, enabling secure and direct access for students.
- **Graduate Teaching Assistant, George Mason University(Aug 2024-Present)**
 - Assisted in a course that teaches server-side scripting using Node.js with emphasis on secure coding practices.
- **R&D Intern -Centre for Distributed Computing, Jadavpur University(Feb 2024-July 2024)**
 - Designed and tested Moving Target Defense (MTD) strategies for Software-Defined Networking (SDN), improving resilience against targeted attacks.
- **R&D Intern -Centre for Digital Transformation and Innovation(Oct 2023-Feb 2024)**
 - Worked on computer vision research, contributing to the development of innovative solutions.
- **Research Intern -George Mason University(July 2023-Aug 2023)**
 - Conducted novel research on network intrusion detection rule optimization using genetic algorithms under Dr. Massimiliano Albanese, which resulted in a paper in SECRIPT 2024.

- **Penetration Tester Trainee -Techgyaan Technologies(Dec 2021)**

Identified and documented 30+ vulnerabilities in sandbox systems, leading to enhanced mitigation protocols.

- **Technical Content Writer for DataFlair(July 2021-Sept 2021)**

Authored articles on emerging Java technologies, helped increase site traffic by SEO.

Managerial Roles:

- **Hack-X Cybersecurity Society Founding Member.** Organized CTF competitions, seminars, and other activities related to cybersecurity.
- **Placement Cell DUK core team Member.** Roles include management of various placement activities.

PUBLICATIONS AND CONFERENCES

Towards AI-Driven Human-Machine Co-Teaming for Adaptive and Agile Cyber Security Operation Centers TOIT

ACM Transactions on Internet Technology

2026

- Proposed an AI-driven human-machine co-teaming paradigm in which LLM-based agents act as “AI apprentices” to assist SOC analysts with threat intelligence, alert triage, and incident response.
- Explored mechanisms for capturing and operationalizing analysts’ tacit knowledge through continuous human-AI interaction, enabling AI agents to adapt to SOC-specific workflows and operational context.
- Demonstrated the feasibility of the proposed paradigm through a preliminary case study conducted in collaboration with a real-world SOC.

<https://doi.org/10.1145/3819819>

Optimizing IDS Rule Placement via Set Covering with Capacity Constraints COSE

Computers & Security

2025

- Formulated IDS rule deployment across multiple detectors as a capacity-constrained set covering problem to minimize rule deployments while maintaining coverage of known network vulnerabilities.
- Developed a priority queue-enhanced backtracking algorithm that incorporates IDS inspection capacity and statistical variations in network traffic to improve resilience under high-traffic conditions.
- Evaluated the approach using CSE-CIC-IDS2017 and a simulated multi-IDS environment, demonstrating improved vulnerability coverage and fewer uninspected packets compared with uniform rule deployment.

<https://doi.org/10.1016/j.cose.2025.104748>

Improving the Efficiency of Intrusion Detection Systems by Optimizing Rule Deployment Across Multiple IDSs SECRYPT

21st International Conference on Security and Cryptography

2024

- Proposed an attack graph-based framework for strategically deploying intrusion detection rules across multiple IDSs based on network topology, vulnerability information, and attack paths.
- Formulated rule placement as an optimization problem that maximizes the number of detectable attack paths while respecting IDS capacity and network traffic constraints.
- Developed a genetic algorithm-based heuristic to identify efficient rule placement strategies and balance detection coverage with computational load across IDS instances.

<http://dx.doi.org/10.5220/0012814500003767>

Ensemble Learning And its Application in Spam Detection ICCECE

International Conference on Computer, Electrical & Communication Engineering

2023

- Investigated heterogeneous ensemble learning for email spam detection by combining Support Vector Machine (SVM), Multinomial Naive Bayes (MNB), Random Forest (RF), and Decision Tree (DT) classifiers.

- Evaluated different combinations of the four classifiers using a max-voting ensemble strategy to reduce classification errors produced by individual machine learning models.
- Identified the SVM, MNB, and DT ensemble as the best-performing model combination among the evaluated configurations.

<https://ieeexplore.ieee.org/document/10085378>

SKILLS

AI & Data Science: Machine Learning, Deep Learning, NLP, Ensemble Learning, Neural Networks, GenAI, AI Agents

Cybersecurity Tools & Techniques: Intrusion Detection Systems (IDS), Penetration Testing, Vulnerability Assessment, Ethical Hacking, Moving Target Defense, SDN Security, MITRE ATT&CK Framework, Kali Linux Tools

Programming & Frameworks: Python, C++, Java, Flask, Node.js, JavaScript, MySQL, GIT, Streamlit, Gradio

Networking: Network Troubleshooting, Protocol Debugging, Network Simulation(GNS3), Routing, Switching

DevOps & Deployment: Server Deployment, Linux Server Administration, Internal Network Hosting, API Integration, Scripting for Network Monitoring

ACHIEVEMENTS AND CERTIFICATIONS

IEEE ComSoc SNTC Student Competition 2025

- Gold Award in the international proposal writing competition, where there were participants from all around the world.

Outstanding Academic Performance Award 2024

- Awarded for my performance in academics and extracurricular activities.

CCNAv7.0 from CISCO Academy 2024

Smart India Hackathon Internals, First runner-up 2023

CTF 1.0 2022

- First Runner-up Organized by KSAAC https://ksaac.duk.ac.in/event_past.php

Cybersecurity and Ethical Hacking Course 2021

- Cyber Security and Ethical Hacking for Beginners and Advanced from GUVI Geek.

IIT Madras Certified Python Course 2021

- Completed an in detailed course on python programming under distance learning by IIT Madras.